CMP450: Object-Oriented Software Engineering

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IntelliRide: Heuristic-Based Ride-Sharing & Carpool Matching System

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1. Literature Review

Bruglieri et al. 's (2011) paper demonstrates a carpooling service designed for universities and demonstrates important features that should be considered for implementation of a carpooling/ridesharing service such as cost estimations and delivery of ride schedules. Since the paper goes over a solution similar in scope to the proposed IntelliRide Project it is an important resource in determining the requirements as it helps demonstrate the needs of similar stakeholders.

Mitropoulos et al.'s (2021) study goes over a systematic review of several ride sharing services and analyzes data to determine factors that would cause a user to use a ride sharing service. Some of the factors covered include sociodemographic factors such as age, education level, trip purpose and income. Other factors include system factors such as security, availability, and flexibility. This analysis is important to use in helping to determine the requirements, both functional and nonfunctional, for the IntelliRide Project.

Shaheen, Chan, and Gaynor (2016) study the behavioral and operational factors affecting shared mobility services, including ride sharing/carpooling. Their findings highlight the importance of flexibility, reliability, and user transparency in encouraging participation. The study supports the need for a transparent matching approach in IntelliRide to ensure student confidence in the system.

Agatz et al. (2012) presents a comprehensive review of optimization and matching techniques used in ride sharing systems, while focusing on route similarity and time compatibility. The study discusses heuristic and algorithmic approaches for efficiently matching drivers and passengers while mitigating detours and delays. This work is relevant to the IntelliRide Project as it provides a strong theoretical foundation for using heuristic based matching instead of opaque machine learning models.

Morency (2007) investigates carpooling behavior and identifies time constraints and trip synchronization as major barriers to successful ride sharing. The study emphasizes that matching users with similar schedules is critical for an efficient system. This directly aligns with IntelliRide's focus on time based schedule aware matching for students with fixed class timetables.

2. Problem Domain

In light of the constant rise in fuel prices, traffic, parking, and environmental pollution, it is safe to say that there are a number of challenges that arise in the field of urban transportation, particularly for university students. Ride-sharing and carpooling can certainly play a role in helping mitigate these challenges, but existing systems that have been put in place have a number of rigid routes and limited or vague matching criteria, which leads to a level of mistrust among the users of such a system.

It appears that a number of solutions have been put in place, especially in the field of machine learning, without a clear explanation of how these matches are made. This leads to a lack of transparency, which can result in a level of mistrust among users, and also makes it more difficult to maintain such a system. A student focused ride sharing system can allow for better matching and heuristic evaluation, making the pool of possible shareable rides significantly larger.

It is therefore necessary to have a clear and transparent ride-sharing system that is able to intelligently match student drivers traveling in similar directions, taking into consideration the constraints of distance, trust, and time.

Problem: Current ride sharing systems fail to capture the most special market, which is of university students, making inefficient use of concise scheduling systems of university students.

Proposed Solution: A heuristic-based approach to ride-sharing and carpool matching that transparently ranks drivers and passengers based on their concise university schedule, distance, time, and trust scores criteria.

3. Product Vision

Our proposed system, IntelliRide, is an intelligent ride-sharing and carpool matching system that will be developed with the objective of:

- Matching drivers and passengers traveling in the same/similar direction according to personal/university schedules.
- Providing transparent and explainable matching scores.
- Using a heuristic-based matching approach instead of vague ML models.
- Building a system that encourages trust and safety through a feedback system.

Key Requirements:

- Users will be able to register as either Drivers or Passengers.
- Passengers will be able to request a ride to their campus/home at a desired time..
- Drivers can create a trip with a route, time, and available seats.

- The system will provide a ranked list of matches with explanations of the scoring system.
- A matching engine will use the following criteria to evaluate potential matches between drivers and passengers:
    - Distance Similarity
    - Concise University Schedule
    - User Trust and Rating Scores
    - Time Compatibility

4. Difference in Proposed Approach

Unlike general purpose ride sharing platforms that operate in open and highly variable environments, our ridesharing application is designed specifically for a university focused ecosystem. Allowing for easy contrasting of matching schedule formats, this will enable more accurate heuristic based matching, particularly in terms of time compatibility and route overlap without relying on ML models. It will also allow for enhanced trust and safety through verified student identities and shared institutional affiliation.

5. Stakeholders

The primary stakeholders for the ride-sharing application will be the drivers and the passengers. The solution needs to cater to both the people driving to make sure it is easy to use, and passengers need features that ensure safety and ease of identification of drivers.

List of Stakeholders:

- Drivers (offer rides, reliable time and destination display, check seat availability)
- Passengers (request destination, time, driver identification)

- System Administrators/Managers (handle users, trips, and platform policies)
- Development Team (design, build, and deploy system)
- Universities/Companies (potential adopters for carpooling environments)
- Government Entities (RTA)

Bibliography

Bruglieri, M., Ciccarelli, D., Colorni, A., & Luè, A. (2011). PoliUniPool: a carpooling system for universities. *Procedia - Social and Behavioral Sciences, 20*, 558–567. https://10.1016/j.sbspro.2011.08.062

Mitropoulos, L., Kortsari, A. & Ayfantopoulou, G. A systematic literature review of ride-sharing platforms, user factors and barriers. *Eur. Transp. Res. Rev.* **13**, 61 (2021). https://doi.org/10.1186/s12544-021-00522-1

Agatz, N., Erera, A., Savelsbergh, M., & Wang, X. (2012). Optimization for dynamic ride-sharing: A review. *European Journal of Operational Research*, *223*(2), 295–303. https://doi.org/10.1016/j.ejor.2012.05.028

Shaheen, S. A., Chan, N. D., & Gaynor, T. (2016). Casual carpooling in the San Francisco Bay Area: Understanding user characteristics, behaviors, and motivations. *Transport Policy*, *51*, 165–173. https://doi.org/10.1016/j.tranpol.2016.01.003

Morency, C. (2006). The ambivalence of ridesharing. *Transportation*, *34*(2), 239–253. https://doi.org/10.1007/s11116-006-9101-9